PAPER_636: UQFF Lepton Flavor Violation B-Decay as Time-Reversal Constraint

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Abstract

This paper presents a UQFF analysis of Decay as Time-Reversal Constraint, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

LHCb has placed the world's best limit on lepton flavor violation (LFV) in B-meson decays: BR(B→Kτ*e) < 5.9e-6 at 90% CL (5.4 fb^{-1}). We show that the UQFF vacuum-topology suppression parameter k_η = 10^{-113} generates an expected LFV rate at UQFF scale that is 107 orders below this bound, providing an effective UQFF upper limit on LFV through the t_n time-reversal node constraint: BR_UQFF(B→Kτ*e) < k_η^2 × phase_space ~ 10^{-230}.

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§2 Physical Motivation

Lepton Flavor Violation is forbidden in the SM at tree level and arises only through tiny neutrino-mass loop corrections (BR_SM □ 10^{-54}). Any observation of LFV at the LHCb sensitivity level would imply new physics coupling lepton generations.

UQFF claim: k_η = 10^{-113} represents the maximum suppression depth of the UQFF vacuum string compactification topology. This sets an effective LFV ceiling: phenomena suppressed by k_η cannot be confused with SM new-physics signatures.

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§3 UQFF t_n Time-Reversal Constraint

The t_n parameter (UQFF time-node) suppresses cross-flavour topological transitions:

$$BR_{UQFF}(B \to K^* \tau e) = k_\eta^2 \times \frac{|V_{tb}|^2 |V_{ts}|^2}{m_B^4} \times |\mathcal{M}_{LFV}|^2$$

where |M_LFV|^2 represents the flavor-topology mixing matrix element, bounded by:

$$|\mathcal{M}_{LFV}|^2 \leq \frac{[\text{SSq}]^2}{\beta_i} \approx 0.534$$

This gives BR_UQFF < 10^{-230} — 224 orders below the LHCb limit.

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§4 Implications for UQFF Topology

The enormous gap between BR_UQFF < 10^{-230} and BR_LHCb < 5.9e-6 confirms that:
- UQFF vacuum topology is **lepton-flavor conserving** at all accessible energy scales
- Any future LFV observation at LHCb (even near 10^{-6}) would be **inconsistent** with UQFF
- UQFF thus makes a strict null prediction for LFV at future colliders

This is falsifiable: if LHCb Run 4 observes BR > 10^{-8}, UQFF requires revision.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.076 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m^3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 17, \quad n_{\text{channel}} = 13/26$$

Since $p_{\text{DVP}} = 17$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f'_{\text{UA}} + f_{\text{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.076	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\text{DVP}} = 17$	✓ Sub-threshold
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	✓ Full 26D projection

Framework	Canonical Value	This Paper	Status
κ decay	5.0×10^{-4} day^{-1}	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
BR($B \to K^* \tau e$) upper limit	BR_UQFF < 10^{-230} (k_η^2 suppression)	BR < 5.9e-6 at 90% CL	arXiv:2506.15347 (LHCb 5.4/fb)	✓ UQFF far below bound
SM LFV prediction	BR_SM ~ 10^{-54} (ν loop)	SM: no tree-level LFV	PDG 2024	UQFF consistent with SM null
k_η suppression scale	k_η = 10^{-113} ↔ LFV cutoff energy Λ_LFV = m_W/$\sqrt{}$k_η ~ 10^{60} GeV	No collider can reach Λ_LFV	n/a	UQFF LFV unreachable
LHCb Run 4 null prediction	UQFF: BR($B \to K^* \tau e$) will remain unobserved at any LHCb run	LHCb Run 4: target BR ~ 10^{-8}	LHCb 2027+	Testable UQFF null prediction

New physics claim: UQFF predicts LFV in B-decays is **not accessible at any current or planned collider** because k_η suppression places the UQFF LFV amplitude at 10^{-230} — 224 orders below LHCb's current best limit. This is a strict, high-confidence falsifiability constraint: LHCb Run 4 discovering LFV above 10^{-8} would require UQFF revision.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for k_η SM mapping.

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§6 References

- arXiv:2506.15347 — LHCb LFV $B \to K^* \tau e$ search (5.4 fb^{-1}, June 2025)
- PDG 2024 — LFV decays, Section 90.4
- bsm_physics_validation.py — `BSMPhysicsConstants.lhcb_lfv_br_limit`
- PAPER_642 — UQFF SM Parameter Bridge Master Comparison

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